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Learning Metric Features for Writer-Independent Signature Verification using Dual Triplet Loss

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***Abstract*—Handwritten signature has long been a widely accepted biometric and applied in many verification scenarios. However, automatic signature verification remains an open research problem, which is mainly due to three reasons. 1) Skilled forgeries generated by persons who imitate the original writing pattern are very difficult to be distinguished from genuine signatures. It is especially so in the case of offline signatures, where only the signature image is captured as a feature for verification. 2) Most state-of-the-art models are writer-dependent, requiring a specific model to be trained whenever a new user is registered in verification, which is quite inconvenient. 3) Writer-independent models often have unsatisfactory performance. To this end, we propose a novel metric learning based method for offline writer-independent signature verification. Specifically, a dual triplet loss is used to train the model, where two different triplets are constructed for random and skilled forgeries, respectively. Experiments on three alphabet datasets — GPDS Synthetic, MCYT and CEDAR — show that the proposed method achieves competitive or superior performance to the state-of-the-art methods. Experiments are also conducted on a new offline Chinese signature dataset — CSIG-WHU, and the results show that the proposed method has a high feasibility on character-based signatures.**

I. Introduction

Handwritten signature is one of the most common biometric features for verification in real life. An expert can distinguish between the genuine and forgeries based on the continuity of lines, joined-up writing patterns, proportion of components and so on.

The signature verification systems fall into two categories: Online (dynamic) and Offline (static) verification. In the offline setting, the signature is acquired after the signing process is completed and usually represented as a digital image. In this case, only the shape of the signature is captured as a feature for verification [1], [2]. In the online setting, however, a special device is used to capture the user's signature. The data is collected as a sequence over time, containing dynamic information such as pressure, azimuth, pen up and down, etc [2].

This paper deals with the offline setting, which is more difficult than the online setting, and is more closer to real life scenario. Traditionally, efforts in this field were largely devoted to finding good feature descriptors using handcrafted feature extractors [2]. But in recent years, there have seen an increasing interest in deep learning models [3], [4], that learn feature representations from raw data, e.g., scanned images, without handcrafted feature extractors.

Based on deep learning, there are generally two types of models: writer-independent (WI) models and writer-dependent (WD) models. The former refers to a single model used to verify signatures from any user. The latter means each user requires a specific model to be trained for verification. The writer-independent verification is usually more feasible in real-life applications. For example, if we were to design a signature verification system for the banking industry, it would be very expensive and inefficient if a specific model needs to be trained and fine-tuned for each newly registered customer.

Traditionally, writer-independent models usually use handcrafted feature extractors, which did not report satisfactory accuracy in some benchmark datasets. Souza et al. [5] proposed a writer-independent approach using CNN features in 2018. Their models achieved quite good results, but relied on pre-trained CNN models and were not trained in an end-to-end manner.

In 2016, Rantzsch et al. [6] first introduced metric learning into this field for writer-independent verification. However, their experiments were conducted on only one dataset and lacked comparison with other works. The method we propose in this paper uses the dual triplet loss. The experiments are conducted on four datasets: GPDS Synthetic dataset [7], CEDAR dataset [8], MCYT dataset [9] and our own Chinese signature dataset: CSIG-WHU. The model presented in our experiments is called SigCNN. We also compare our method with other feature-learning algorithms and writer-dependent models.

Our main contributions in this paper are as follows: 1) We propose a novel writer-independent method for offline signature verification. Experiments show that our method considerably surpassed state-of-the-art models in some benchmark datasets. 2) The model we propose is writer-independent and trained in an end-to-end fashion. This means metric features it learned can be transferred to many real-life applications once been well-trained and fine-tuned, without any re-processing or re-training. 3) We conduct our experiments on our own Chinese datasets, CSIG-WHU, and three alphabet datasets. This could unveil differences in features between character-based and alphabet-based languages, and the feasibility of training one model for both.

The remaining part of this paper is organised as follows. Section II briefly reviews the related work. Section III introduces our proposed method. Section IV reports our experiments and results. Sections V concludes this paper by evaluating our method and briefly discussing possible future work.

II. Related Work

Research on offline signature verification can be traced back to 1970s. Over the decades, approaches have been proposed from different perspectives, as summarised in [10], [11] and [2].

This problem can be formulated as learning a model that tells forgeries from the genuine given a set of genuine signatures. Forgeries have two types: random forgery, where a forger uses his or her signature to impersonate the other individual; and skilled forgery, which refers to signatures imitated from genuine ones [12].

Similar to pattern recognition in other applications, such as 3D super-alloy image segmentation [13] or ancient paintings dating [14], research in offline signature verification also aims to find a good feature descriptor for extraction.

In this section, we briefly review previous work in three aspects: handcrafted feature extractor, public datasets and deep learning.

A. Handcrafted Feature Extractor

Before deep learning models attracted increasing attention of the research community, a large amount of feature extractors has been proposed for this problem, from geometric feature descriptors [15]–[17], directional feature descriptors such as Directional-PDF (Probability Density Function) [18] and pyramid histogram of oriented gradients (PHOG) [19], to texture feature descriptors such as Local Binary Patterns (LBP) [20], [21] and Gray Level Co-occurrence Matrix (GLCM) [22], [23]. There are also some pseudo-dynamic feature extracting algorithms based on studies of graphometry, which restores dynamic information such as speed, continuity and uniformity from the static signature image [24].

B. Public Datasets

Research in this field, especially those involving deep learning, requires large public datasets. Most public datasets available today follow similar process to acquire signature image. Users are required to provide their genuine samples on a paper with many cells. Forgers then are offered those genuine sample and required to imitate the signatures one or more times. After those forms are collected, they are scanned and preprocessed. In some datasets, factors such as age, gender, form of pen, etc. are also taken into account. The GPDS960Gray [25] used to be the largest offline signature dataset in the research community, but is no longer available to the public because of privacy legislations. Besides GPDS960Gray, other datasets are much smaller ones, with CEDAR and MCYT two of the most commonly used. Some other datasets also used algorithms to automatically generate genuine or forged signatures. In this case, they can be very large in size but might not be suitable for training models targeting real signatures. Among these synthetic datasets, GPDS Synthetic dataset [7] is the largest and most recently used in the literature.

C. Deep Learning

After deep learning were introduced into this field, early work usually used private dataset and did not report much success. Ribeiro et al. [26] used Restricted Boltzmann Machines (RBMs) to learn features from signature images but only showed visual appearance of the weights without testing those features for verification. Khalajzadeh [27] used Convolutional Neural Network (CNN) for Persian signature verification, but only considered random forgeries in their work.

To distinguish between genuine signatures and skilled forgeries, there are two kinds of approaches in recent literature, as summarised in [2]: 1) train writer-dependent classifiers based on pre-learned writer-independent features. 2) learn feature representation at once for a write-independent system, usually using metric learning.

Hafemann et al. [28] proposed a method that learns a writer-independent feature representation function $\sigma(x)$ using CNN on a development set $\mathcal{D}$, which later is used as feature extractor for training writer-dependent classifiers on the exploitation set ε. In later work [12], the authors also proposed a multi-task framework, where CNN was trained jointly on genuine signatures and skilled forgeries. Zhang et al. [29] proposed a feature learning method based on Generative Adversarial Networks (GAN) [30]. In this case, a generator (that learns to generate signature images) and a discriminator (that learns to discriminate between real signatures and generated ones) are trained on a subset of users. After the training process is complete, the authors used convolutional layers of the discriminator as the feature extractor.

Rantzsch et al. [6] proposed a writer-independent approach using metric learning, where distances between signatures were learned for verification. In the training process, tuples that consist of three signatures are fed into the neural network: (X_r, X_+, X_-), where X_r refers to a reference sample, X_+ a genuine sample, and X_- a forgery (either a random one or a skilled one). The neural network is trained to minimise the distance between X_r and X_+, while maximising the distance between X_r and X_-. After training, signatures with a distance to reference sample under a certain threshold are classified as genuine ones, and those over the threshold as forgeries.

Recently there are also researches that introduce meta-learning into this field. Hafemann et al. [31] proposed a model in 2019 that uses a meta-learner to leverage data from multiple tasks for learning. In their model, writer-dependent classifiers for each user are seen as tasks, and an extended version of Model-Agnostic Meta-Learning (MAML) [32] is used for training.

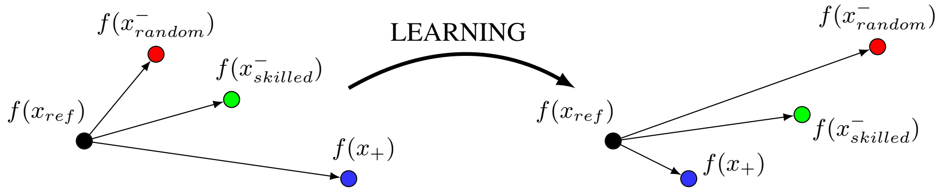

Fig. 1: When training using dual triplet loss, we hope to make $f(x_+)$ closer to $f(x_{ref})$ than both $f(x^-_{random})$ and $f(x^-_{skilled})$. Also, $f(x^-_{random})$ should be intuitively farther from $f(x_{ref})$ than $f(x^-_{skilled})$.

III. Proposed Method

A. Dual Triplet Loss

The basic idea of our proposed metric learning method is to obtain feature embeddings where distances between reference and genuine samples are larger than those between reference and forgery samples. It can be seen as an extended version of triplet loss. For more detailed explanations of triplet loss, we refer our readers to [33] and [34]

We denote the embedding function as $f(x) \in \mathbb{R}^d$. It projects the input image x into a d-dimensional hypersphere within an Euclidean space, *i.e.* $\|f(x)\|_2 = 1$. In our model, this embedding function is a Convolutional Neural Network (CNN). In this case, the feature embeddings of the reference sample are given by $f(x_{ref})$, genuine samples $f(x_+)$, random forgeries $f(x^-_{random})$ and skilled forgeries $f(x^-_{skilled})$.

Here we want to ensure $f(x_+)$ is closer to $f(x_{ref})$ than both $f(x^-_{random})$ and $f(x^-_{skilled})$. Moreover, $f(x^-_{skilled})$ should be intuitively closer to $f(x_{ref})$ than $f(x^-_{random})$, as random forgeries should be much easier to recognise. This process is visualised in Fig. 1.

We use Euclidean distance and thus we want,

$$\|f(x_{ref}) - f(x_+)\|_2^2 + \alpha_1 < \|f(x_{ref}) - f(x^-_{skilled})\|_2^2 \quad (1)$$

$$\|f(x_{ref}) - f(x_+)\|_2^2 + \alpha_2 < \|f(x_{ref}) - f(x^-_{random})\|_2^2 \quad (2)$$

$$\forall\ (f(x_{ref}),\ f(x_+),\ f(x^-_{random}),\ f(x^-_{skilled})) \in \mathcal{T}$$

where α_1 and α_2 are margins between genuine and forgery embeddings, and intuitively we have $\alpha_1 < \alpha_2$. $\mathcal{T}$ is the set of all possible dual triplets in the training set. It can also be seen as quadruplets, but we refer to them as dual triplets, in order to remove ambiguity.

Now if we see random and skilled forgeries respectively, the loss functions of the two can be computed as:

$$\begin{aligned}\mathcal{L}_{skilled} &= [\ \|f(x_{ref}) - f(x_+)\|_2^2 - \\ &\quad \|f(x_{ref}) - f(x^-_{skilled})\|_2^2\ +\ \alpha_1\ ]_+\end{aligned} \quad (3)$$

$$\begin{aligned}\mathcal{L}_{random} &= [\ \|f(x_{ref}) - f(x_+)\|_2^2 - \\ &\quad \|f(x_{ref}) - f(x^-_{random})\|_2^2\ +\ \alpha_2\ ]_+\end{aligned} \quad (4)$$

Thus the final loss function can be computed as the weighted sum of $\mathcal{L}_{skilled}$ and $\mathcal{L}_{random}$:

$$\mathcal{L} = \sum_{\mathcal{T}} \lambda\ \mathcal{L}_{skilled} + (1-\lambda)\ \mathcal{L}_{random} \quad (5)$$

where λ is a constant between 0 and 1.

However, using loss functions above can be computationally expensive during training. We attempt to reduce computational cost by employing the batch hard strategy and soft margin.

B. batch hard triplet mining & soft margin

In order to compute the dual triplet loss function, we need to find out a mining strategy that selects valid triplets violating constraints given by Eq.1 and Eq.2. In our training protocol, for every reference sample x_{ref}, we only select the online triplets $(x_+, x^-_{random}, x^-_{skilled})$ satisfying:

$$argmax\ _{x_+}\|f(x_{ref}) - f(x_+)\|_2^2$$

$$argmin\ _{x^-_{random}}\|f(x_{ref}) - f(x^-_{random})\|_2^2$$

$$argmin\ _{x^-_{skilled}}\|f(x_{ref}) - f(x^-_{skilled})\|_2^2$$

This means we only focus on the hardest genuine samples and hardest forgery samples (both random and skilled) within every mini-batch, and all invalid or easily satisfied triplets will not be taken into account during training. In this way the computational cost can be significantly reduced and the model is expected to converge much faster.

We also noticed that the hinge functions in Eq.3 and Eq.4 have a hard cut-off at the margins α_1 and α_2 that are enforced between genuine and forgery samples. This will cause the model to have zero gradients when the distance between genuine and forgery embeddings exceeds a certain point, or the distance between genuine embeddings are small enough. However, in the offline signature verification, we hope to pull together signatures from one user, and push away signatures from different users as much as possible. Therefore we replace the hinge function with a smooth approximation proposed in [34]:

$$\ln(1 + \exp(\cdot))$$

Thus we rewrite the loss functions as follows:

$$\begin{aligned}\mathcal{L}_{skilled} &= \ln(1 + \exp(\max_{x_+ \in \mathcal{T}_{batch}} \|f(x_{ref}) - f(x_+)\|_2^2 \\ &- \|\min_{x^-_{skilled} \in \mathcal{T}_{batch}} f(x_{ref}) - f(x^-_{skilled})\|_2^2\ ))\end{aligned} \quad (6)$$

SigCNN Architecture		
layer	size	kernel param
input	$1 \times 150 \times 220$	
conv1	$96 \times 11 \times 11$	stride = 4, pad = 0
pool1	$96 \times 3 \times 3$	stride = 2
conv2	$256 \times 5 \times 5$	stride = 1, pad = 2
pool2	$256 \times 3 \times 3$	stride = 2
conv3	$384 \times 3 \times 3$	stride = 1, pad = 1
conv4	$384 \times 3 \times 3$	stride = 1, pad = 1
conv5	$256 \times 3 \times 3$	stride = 1, pad = 1
pool3	$256 \times 3 \times 3$	stride = 2
fc1	2048	
fc2	2048	
fc3	128	
L2	128	

TABLE I: The SigCNN arhitecture used in our experiments. All sizes are described in $height \times width \times filters$. We used batch normalisation after each layer, and rectified linear units (ReLU) as the activation function.

$$\begin{aligned}\mathcal{L}_{random} &= \ln(1 + \exp(\max_{x_+ \in \mathcal{T}_{batch}} \|f(x_{ref}) - f(x_+)\|_2^2 \\ &- \| \min_{x^-_{random} \in \mathcal{T}_{batch}} f(x_{ref}) - f(x^-_{random})\|_2^2 \ + \ \alpha))\end{aligned} \tag{7}$$

$$\mathcal{L} \ = \ \sum_{\mathcal{T}_{batch}} \lambda \ \mathcal{L}_{skilled} + (1 - \lambda) \ \mathcal{L}_{random} \tag{8}$$

where $\mathcal{T}_{batch}$ represents all anchors within every mini-batch during training, $\mathcal{T}_{hard}$ represents hardest triplets with regards to the current reference anchor. α in Eq.7 determines how farther we want random forgeries to be from the genuine than skilled forgeries.

IV. Experiment

A. Model Architecture

The model used in our experiments is inspired by SigNet proposed by Hafemann et al. in [12]. We call it SigCNN. Table I specifies the architecture of the SigCNN. This network projects all input images into a 128-dimensional hypersphere.

When testing our models, we first calculate the centre of all embeddings of reference samples (note that the centre point is also L2-normalised and thus lies on the 128-dimensional hypersphere), and the mean distance of all reference embeddings to the centre. We then compare the distance between the centre and embeddings of questioned samples, with the mean distance. This means in our test we also use relative distance, similar to the loss function in our training process.

B. Datasets

We conducted our experiments on four datasets: GPDS Synthetic dataset, CEDAR dataset, MCYT dataset, and our own Chinese dataset CSIG-WHU.

Fig. 2: A sample user of our CSIG-WHU. The top two rows are genuine signatures, and the bottom two rows are forgeries from two different forgers, respectively. In our dataset, pens of different thickness are used.

The GPDS Synthetic dataset is a synthetic dataset that consists of 4000 synthetic users. Each user has 24 genuine signatures and 30 skilled forgeries.

The MCYT and CEDAR are much smaller datasets. The MCYT has 75 users, each with 15 genuine signatures and 15 skilled forgeries, while the CEDAR contains 55 users, each with 24 genuine signatures and skilled forgeries.

We also tentatively conducted our experiment on the Chinese dataset, CSIG-WHU. This dataset has 60 users and a total of 2815 offline Chinese signatures. Each user has 14.4 genuine signatures and 32.5 skilled forgeries on average. In CSIG-WHU, All signatures were collected using pens of different thickness, and writers were encouraged to slightly change their writing pattern or even signing position (seated or standing). In this way we hope to make our data as close to wild data as possible.

C. Training Protocol

We first trained our model on GPDS Synthetic dataset. We selected the first 3000 users as the development set, the last 800 users as the exploitation set, and the remaining 200 users as the validation set. On MCYT and CEDAR dataset, we transferred our pre-trained model directly, and then finetuned our model using the first 5 users.

In our training process, we pre-processed all signature images by reading them in grayscale, and then performing bitwise negation on each pixel, which was initially represented as an 8-bit integer. We then resized all images into 220×150 (width$\times$height) tensor.

We set $\alpha = 1$ and $\lambda = 0.9$. We used RMSprop optimiser and the initial learning rate was set to 0.0001. We also used early stopping, which means we stopped training when the performance on validation set started dropping.

In order to compare our work with former state-of-the-art, we trained SigNet-F model on GPDS Synthetic dataset following the same split of dataset, and the same protocol as specified in [12].

D. Results

We consider the following metrics to evaluate the performance:

TABLE II: Performance of SigCNN and SigNet-F [12] on GDPS Synthetic dataset (%)

Model	**#Refs**	**FRR**	**FAR$_{\text{random}}$**	**FAR$_{\text{skilled}}$**	**AUC**	**EER**	**EER$_{\text{user}}$**
SigNet-F	4	55.31(±1.40)	0.14(±0.08)	8.10(±0.30)	86.35(±0.40)	20.68(±0.52)	16.25(±0.27)
	8	38.19(±0.60)	0.13(±0.08)	6.84(±0.27)	88.99(±0.22)	18.24(±0.21)	13.48(±0.34)
	12	27.63(±0.33)	0.11(±0.03)	5.44(±0.40)	91.43(±0.22)	15.99(±0.33)	12.10(±0.28)
SigCNN	4	36.61(±1.14)	0.04(±0.02)	2.53(±0.28)	95.91(±0.20)	10.38(±0.23)	N/A
	8	35.23(±1.05)	0.02(±0.01)	1.71(±0.11)	97.28(±0.08)	8.12(±0.15)	N/A
	12	39.54(±0.92)	0.01(±0.01)	1.21(±0.06)	97.69(±0.08)	7.34(±0.25)	N/A

TABLE III: Comparison with the state-of-the-art on GPDS Synthetic dataset

Reference	Type	#Refs	Method	EER(%)
Soleimani et al. [35]	WD	10	HOG+DMML	12.14
Ferrer et al. [36]	WI	10	LBP+SVM	16.44
Serdouk et al. [37]	WD	10	HOT+AIRSV	16.68
Zhang et al. [29]	WD	10	GAN	14.79
Hafemann et al. [12]	WD	5	$\text{SigNet-F}_{\text{user } \tau}$ (pre-trained)	24.91
Hafemann et al. [12]	WD	12	$\text{SigNet-F}_{\text{user } \tau}$ (pre-trained)	16.98
Hafemann et al. [12]	WD	4	$\text{SigNet-F}_{\text{user } \tau}$	16.25(±0.27)
Hafemann et al. [12]	WD	8	$\text{SigNet-F}_{\text{user } \tau}$	13.48(±0.34)
Hafemann et al. [12]	WD	12	$\text{SigNet-F}_{\text{user } \tau}$	12.10(±0.28)
Ours	WI	4	SigCNN	10.38(±0.23)
Ours	WI	8	SigCNN	8.12(±0.15)
Ours	**WI**	**12**	**SigCNN**	**7.34(±0.25)**

1) Area Under the Curve (AUC): Considering the ROC curve [38] on the test set, we calculate the area under the curve.

2) False Rejection Rate (FRR): The proportion of genuine signatures mistakenly rejected as forgeries.

3) False Acceptance Rate (FAR): The proportion of forgeries mistakenly accepted as genuine signatures. We consider both random forgeries and skilled forgeries and calculate two False Acceptance Rate respectively: $\text{FAR}_{\text{random}}$ and $\text{FAR}_{\text{skilled}}$.

4) Equal Error Rate (EER): To calculate Equal Error Rate (EER) we only consider skilled forgeries, and the EER is obtained when $\text{FRR} = \text{FAR}_{\text{skilled}}$.

We tested all models subject to different number and different choices of reference samples.

It is worth noting that, for writer-dependent models, the reference samples in test sets are used for training writer-dependent classifier, usually a Support Vector Machine (SVM), whilst in our models they are used for calculating metric feature embeddings and the threshold for verification. The memory overhead and computational expense of the latter is significantly reduced compared to the former.

Table II summarised the performance of our SigCNN and SigNet-F on GPDS Synthetic dataset. We found that the SigCNN surpassed the former state-of-the-art by almost all metrics, achieving an EER of 7.34% when there are 12 reference samples available for verification. A comparison between SigCNN and other state-of-the-art is given by Table III, which manifests that our model considerably surpassed all previous methods on this dataset.

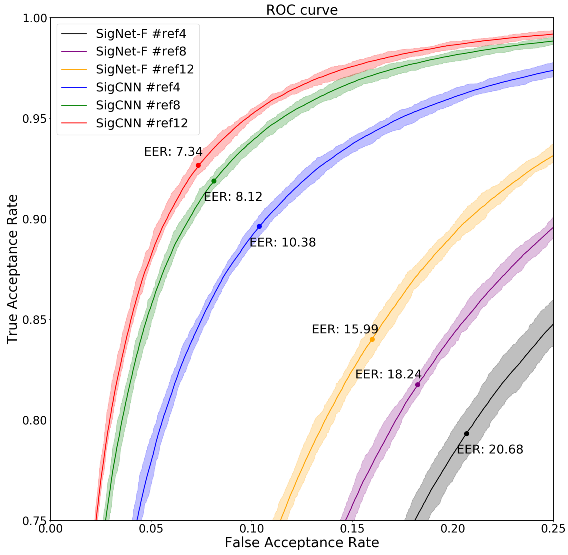

Fig. 3: ROC curve of SigCNN and SigNet-F models proposed by Hafemann et al. [12] on the GPDS Synthetic dataset.

We then directly transferred our model to CEDAR and MCYT dataset, also showing very promising results, as summarised in Table V and Table VI. We noticed that the pre-trained SigCNN is very competitive to former state-of-the-art on CEDAR dataset, and almost closed the problem on CEDAR after finetuning using the first 5 users, hitting nearly 0.0% EER.

Results on MCYT might not seem very impressive, but still quite competitive after finetuning, reporting a best of 12.83% EER. Differences in performance between the CEDAR and MCYT datasets might be because the distribution of GPDS Synthetic dataset is closer to CEDAR than MCYT.

E. Transfer to CSIG-WHU

We also transferred SigCNN to our own Chinese dataset, CSIG-WHU and found that our model has a better capability of generalisation than SigNet-F, as shown in Table IV.

However we found that experiments reported lower performance on CSIG-WHU than CEDAR and MCYT for both

TABLE IV: Performance on CSIG-WHU obtained by models trained on GPDS Synthetic

Model	**#Refs**	**AUC**	**EER**	**EER**$_{\text{user}}$
SigNet-F	4	$78.30(\pm1.11)$	$28.65(\pm1.78)$	$23.30(\pm3.63)$
SigNet-F	8	$82.68(\pm1.45)$	$25.30(\pm1.51)$	$21.28(\pm1.09)$
SigCNN	4	$88.13(\pm1.59)$	$19.94(\pm1.36)$	N/A
SigCNN	8	$90.99(\pm1.34)$	$16.93(\pm1.30)$	N/A

TABLE V: Comparison with the state-of-the-arts on CEDAR dataset

Reference	Type	#Refs	Method	EER
Chen et al. [39]	WD	16	Graph Matching	7.9
Kumar et al. [40]	WI	1	Morphology	11.81
Kumar et al. [41]	WI	1	Surroundness	7.08
Bharathi et al. [42]	WD	12	Chain code	7.84
Guerbai et al. [43]	WD	4	Curvelet transform	8.7
Guerbai et al. [43]	WD	8	Curvelet transform	7.83
Guerbai et al. [43]	WD	12	Curvelet transform	5.6
Hafemann et al. [12]	WD	4	SigNet-F (user τ)	5.92
Hafemann et al. [12]	WD	8	SigNet-F (user τ)	4.77
Hafemann et al. [12]	WD	12	SigNet-F (user τ)	4.63
Hafemann et al. [31]	WI/WD	4	MAML one-class (user τ)	8.27
Hafemann et al. [31]	WI/WD	8	MAML one-class (user τ)	7.07
Ours	WI	5	SigCNN (pretrained)	$8.12(\pm0.98)$
Ours	WI	12	SigCNN (pretrained)	$6.42(\pm1.15)$
Ours	**WI**	**1**	**SigCNN (finetuned)**	$\approx \mathbf{0.0}$

TABLE VI: Comparison with the state-of-the-arts on MCYT dataset

Reference	Type	#Refs	Method	EER
Wen et al. [44]	WD	5	RPF	15.02
Vargas et al. [23]	WD	5	LBP	11.9
Vargas et al. [23]	WD	10	LBP	7.08
Ooi et al. [45]	WD	5	DRT + PCA	13.86
Ooi et al. [45]	WD	10	DRT + PCA	9.87
Soleimani et al. [35]	WD	5	HOG	13.44
Soleimani et al. [35]	WD	10	HOG	9.86
Hafemann et al. [12]	WD	5	SigNet (user τ)	3.58
Hafemann et al. [12]	WD	10	SigNet (user τ)	2.87
Hafemann et al. [31]	WI/WD	5	MAML one-class (user τ)	12.77
Hafemann et al. [31]	WI/WD	10	MAML one-class (user τ)	12.44
Ours	WI	5	SigCNN (pretrained)	$17.24(\pm1.07)$
Ours	WI	10	SigCNN (pretrained)	$15.04(\pm0.93)$
Ours	WI	5	SigCNN (finetuned)	$14.51(\pm1.20)$
Ours	WI	10	SigCNN (finetuned)	$12.83(\pm0.93)$

models, and finetuning using a few users hardly brought any noticeable improvements. This might be due to two reasons:

1) : There are certain differences in features between character-based and alphabet-based signatures. Therefore models pre-trained on alphabet-based signature datasets might not perform satisfactorily on character-based signatures.

2) : Most offline signatures in benchmark datasets are very different from data in the wild. Therefore when we require our subjects to use pens of different thickness, or slightly change their writing pattern, those pre-trained models might hardly be working well.

V. Conclusion

In this paper, we proposed a metric learning method using dual triplet loss for offline signature verification. It is writer-independent and trained in an end-to-end fashion. It is fast, light-weight, convenient, and scalable. Once trained and finetuned, it can be applied to many scenarios without any complications in pre-processing. In our experiments it surpassed the comparison methods on the GPDS Synthetic dataset. It also obtained fairly promising results when applied to other datasets.

However, we found that models pre-trained on benchmark datasets might not report very satisfactory results on our Chinese dataset CSIG-WHU. We also noticed that selecting a good global threshold could be very difficult, as a threshold that delivers good overall results might lead to great variation in performance among users, which means some users might have a very large FRR or FAR whilst these metrics are very satisfactory with regarding to the whole dataset.

Therefore, we believe that, in order to solve these problems, we need more real-signature and character-based data in the wild to finetune our model. We plan to collect more data as close to those in the wild as possible into our CSIG-WHU, and make it public available in the future. We might also consider the feasibility of combining signatures from different languages, either character-based or alphabet-based, into one dataset, and training a model once for universal applications.

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